Juan David Muñoz-Bolaños

Optical System Engineer

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Summary

Physicist and optical engineer with deep expertise in **laser interferometry, wavefront sensing, and precision optical metrology**. Proven track record in end-to-end prototype development, from optical concept and simulation (ZEMAX, Python) through hardware-software integration (Python, C++, LabVIEW) to validated, publication-grade systems. Strong analytical skills and experience transitioning R&D concepts into robust, reproducible measurement setups. Author of three peer-reviewed publications in applied optics and photonics.

Professional Experience

Jan 2022 to Present **Medical University of Innsbruck** Innsbruck, Austria
PhD Researcher, Laser Interferometry & Adaptive Optics

Interferometric Wavefront Sensing: Designed and built a Michelson-type interferometric wavefront sensor (off-axis and phase-stepping with piezo actuators) for adaptive optics microscopy; implemented iterative phase retrieval algorithms and a C++/Python/LabVIEW control stack for scientific cameras and deformable mirrors.

Prototype Development & Validation: Led the end-to-end assembly and commissioning of a custom nonlinear (multiphoton) microscope, including fs-laser integration, galvo-scanner control, and PMT signal acquisition, achieving diffraction-limited resolution.

Physical Modelling & Simulation: Performed systematic ZEMAX tolerance analysis and wave-field simulations (Zernike decomposition) to characterize aberration sensitivity of custom microscope objectives.

Technical Documentation: Published three peer-reviewed articles in ACS Photonics, Biomedical Optics Express, and Applied Sciences, demonstrating whitepaper-level scientific writing.

Jun 2020 to Dec 2021 **Leibniz Institute of Photonic Technology (IPHT)** Jena, Germany
R&D Research Assistant, Photonics & Spectrometer Prototyping

Spectrometer Prototyping: Designed, built, and calibrated a dispersive 1064 nm fiber-probe Raman spectrometer for tissue characterization; published results in Applied Sciences.

Data Acquisition & Analysis: Developed 'SpectraMap' (custom Python package) for supervised ML classification of hyperspectral Raman images, optimizing measurement throughput and evaluation.

Mar 2018 to Mar 2019 **INTECOL S.A.S.** Medellín, Colombia
Junior Machine Vision System Engineer

Optical System Design: Engineered custom illumination (dark-field, bright-field, coaxial) and Halcon pipelines for bottle labeling lines (liquid levels, cap sealing); implemented Cognex VisionPro for automatic part recognition in automotive lines.

R&D-to-Production: Transitioned prototype vision setups to production-ready deployments; performed Gage R&R assessments and feasibility studies for client sign-off.

Technical Competencies

Laser Interferometry & Metrology: Wavefront sensing, phase retrieval, Michelson interferometry, adaptive optics, precision alignment of laser-based measurement systems.

Prototyping & Optical Design: End-to-end system build from concept to validated prototype. ZEMAX/OpticStudio modelling, tolerance analysis, opto-mechanical integration, component qualification (Thorlabs, Edmund Optics, Hamamatsu).

Data Acquisition & Simulation: Python, C++, LabVIEW/FPGA, MATLAB for real-time data capture, signal processing, physical modelling, and instrument control. Experience with JAX (GPU computing).

Technical Documentation: Three peer-reviewed publications (ACS Photonics, Biomedical Optics Express, Applied Sciences). Experienced in writing application notes, technical specifications, and whitepaper-style documents.

Industrial Optics: Halcon & Cognex VisionPro pipelines, custom illumination design, camera integration, Gage R&R, system commissioning on production lines.

Education

Jan 2022 to Present	**Medical University of Innsbruck** *Ph.D. in Adaptive Optics & Microscopy*	Austria
2019 to 2021	**Friedrich Schiller University Jena** *M.Sc. in Photonics*	Germany
2012 to 2018	**University of Cauca** *B.Sc. in Physics Engineering*	Colombia

Publications & awards

Awards: ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional) · German (B2, intermediate/ working proficiency and ongoing) · Spanish (Native)

Soft skills

Collaboration & Communication: Communicated complex technical results through peer-reviewed publications, conference presentations (SPIE Photonics West), and international seminars within cross-functional R&D teams.

Problem Solving: Tackled complex, cross-disciplinary challenges resulting in three peer-reviewed publications and a conference proceeding presented at SPIE Photonics West, San Francisco.

Adaptability: Rapidly integrated into three distinct research environments across three countries (University of Cauca, Leibniz-IPHT Jena, Medical University of Innsbruck), each with different workflows, toolchains, and scientific cultures.

Hobbies

Mountain Biking, Dancing Bachata, Learning Harmonica, Learning German.

References

Prof. Monika Ritsch-Marte Medical University of Innsbruck
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Dr. Christoph Krafft Leibniz Institute of Photonic Technology
Spectroscopy Group Leader
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